Sanjith Shanmugathashan

sanjith1999.github.io shansanjithofficial@gmail.com sanjith-shanmugathashan

DSP/FPGA engineer who takes signal-processing ideas from model to working hardware, and currently leads a team doing the same. Every system I have built has taught me one thing: models are tidy; channels, amplifiers and clocks are not. That gap keeps me happily puzzled, and I keep looking for new problems that sharpen my understanding.

Experience

- **Thakshana Technologies – Secure Communications** Colombo, Sri Lanka
  Team Lead DSP / FPGA (R&D) *May, 2025 – Present*
  - Owned the spread-spectrum waveforms of the DVB-RCS2 based secure TDMA system: set the modulator and demodulator architecture, made the design decisions and split the work across the team. Brought the burst detector up on DDR-backed deep buffering to hold spread bursts, and drove acquisition and synchronization down to -11 **dB SNR**, including the failure modes that surface only at full receiver-chain integration.
  - Designed the RTL **jammer detector** for protected waveforms, operating at up to **30 dB JSR** and built for **120 MSa/s**, with per-hop jammer metrics reported to higher layers and waveform-dependent thresholds under their control. Laid the groundwork it runs on: RMS information carried from the digital front end through the burst detector, and the reporting structure up to higher layers.
  - Built an end-to-end MATLAB model of **Zak-OTFS** over TDL channels, with a linear-complexity equalizer designed to stay implementable. Acquired signals from the energy peak on the delay-Doppler grid, then extended the idea to blind detection using the **Chinese Remainder Theorem**. Took the ∼1 MHz, RRC-shaped chain over the air on an Ettus USRP driven from MATLAB — first in RF loopback, then over antennas. Now designing experiments on link budget, power-amplifier nonlinearity and the in-band distortion it introduces, and channel-emulator validation.
  - Developed a feedback **Kalman equalizer** in MATLAB for **MIL-STD-188-110B** over Watterson-fading HF channels, benchmarked against the figures in the standard inside the integrated channel model. Implemented it in C++ on an Arm Cortex-A53 with headroom to **40 kSym/s** for planned 110C wideband operation, validated that implementation against MATLAB vectors, and supported integration.
  - Led exploratory and bring-up work beyond the main datapath: **DFX** experiments swapping FPGA modules at runtime with millisecond-scale reconfiguration, performance evaluation of a DDR-backed interleaver (reaching ∼**80 Mb/s**), and bring-up of the AFE7444 RF front end over **JESD204B**.
  - Mentor and lead a team of 15 design engineers on RTL development: architectural guidance, code review, and the FPGA/DSP implementation practices the team works to.

  DSP / FPGA Engineer (R&D) *Jan, 2024 – Apr, 2025*
  - Took over design and maintenance of the modulator and demodulator for the DVB-RCS2 MF-TDMA modem, covering **32+ waveforms** and frequency-hopped streams. Designed its core blocks — carrier frequency and phase recovery; turbo FEC decoder — and split the remaining block work across the team, then integrated the chain and validated it over digital loopback. Built the regression setup that keeps it verifiable as updates land, and met the hard requirements: **80 MSym/s** demodulator rate, **100 Mb/s** output rate, $<$**50** μ**s** modulator latency, and backpressure end to end.
  - Led development of the PS–PL data-movement subsystem: a customized AXI-DMA and a PetaLinux driver on hugepage-backed buffers. It carries variable-length packet streams and sustains **758 Mb/s** even at the worst-case 64-byte packet size.
  - Owned RTL design of the **GMSK** and **Narrowband HF Hop** modulators: derived the fixed-point MATLAB model from the floating-point reference to settle quantization, set the top-level plan and core structure the team built to, then handled integration and supported delivery debug.
  - Built two stages of a receive front-end resampling chain in VHDL: a Xilinx FIR IP filter and a hand-written double-rate half-band decimator. Integrated and verified the full chain — fractional resampler and CIC included — resampling a fixed **491.52 MSa/s ADC** stream to any rate from **0.5 to 120 MSa/s**, with timing and AGC gain metadata carried alongside.
  - Designed the Digital Automatic Gain Control for the **DVB-S2** forward-link receiver of the satellite modem in VHDL: a double feedback-loop structure sustaining **120 MSa/s** that holds gain stable against jammers up to **30 dB JSR**, and characterized convergence, threshold and stability across symbol rates alongside the algorithm team.

- **Wadduwage Lab – Harvard University** — Cambridge, MA (Remote)
  Undergraduate Researcher – Computational Imaging — *Aug, 2023 – Jun, 2024*
  - Led the team in simulating a novel 3D computational fluorescence microscope that encodes volumetric information onto a 2D detector using structured illumination generated by a Diffractive Deep Neural Network (D2NN).
  - Utilized the Plug-and-Play reconstruction framework for 3D volumes using 2D denoisers applied along multiple planes (e.g. xy, yz, zx), enabling high-quality 3D reconstructions from compressed temporal measurements.
- **Singapore University of Technology and Design** — Singapore
  Research Assistant Fellow – BLE AoA Localisation — *Jan, 2023 – Aug, 2023*
  - Prototyped a Bluetooth Low Energy (BLE) angle-of-arrival (AoA) based localisation system, focusing on direction estimation and spatial inference.
  - Implemented and evaluated signal processing pipelines for AoA estimation, including phase-based measurements across antenna elements and sensitivity analysis to noise, multipath and hardware impairments.

Education

- **University of Moratuwa** — Moratuwa, Sri Lanka
  BSc (Electronic and Telecommunication Engineering) — *Jan, 2020 – Jun, 2024*
  CGPA : 4.12/4.2; Gold Medalist

Skills

- Programming : ***VHDL***, ***MATLAB***, *Embedded C/C++, Git*
- Tools : ***Xilinx Vivado***, *Questa Simulator,* ***PetaLinux***, *Altium*

Awards/ Achievements

- **Robofest - SINDiB** — *Sep, 2023*
  Sri Lanka Institute of Information Technology (SLIIT) - 2nd runners-up
- **PCB Design & Fabrication** — *Dec, 2020*
  IEEE Student Branches of University of Peradeniya & Open University of Sri Lanka - Won

Certifications

- Zak-OTFS and Delay-Doppler Processing for Next-Gen Communications — *IEEE: Oct, 2025*
- Designing with AXI DMA and using AXI DMA under Linux — *Green-Electrons: Apr, 2025*
- Designing FPGAs Using Vivado Design Suite 1 — *AMD: Jul, 2024*
- Designing with VHDL — *AMD: Jul, 2024*